

Venkata Satish Kumar Bottu BSc,MCA,MBA (PM)

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With 20 years of leadership in **Medical devices, Railway signaling, software applications, and embedded/IoT systems**. Proven track record delivering **full product lifecycle** results from concept and regulatory readiness through global commercialization driving **multi-million revenue growth** and operational excellence.

Expert in **Embedded systems, SaMD**, connected device ecosystems, **cloud & mobile solutions**, and real-time embedded platforms. Skilled in leading cross-functional engineering and product teams while ensuring **regulatory, cybersecurity, and quality** compliance (ISO 13485, IEC 62304, FDA, MDR, CDSO, HIPAA, GDPR, CENELEC, RDSCO).

Recognized for **innovation**, scalable process development, stakeholder alignment, and execution excellence that consistently improves **customer outcomes and business performance**.

EXPERIENCE

Engineering Director, Spacelabs Healthcare (OSI Systems) - 2016-2025

- Led end-to-end new and sustenance product development with global teams collaboration for 4+ regulated medical device platforms (embedded + mobile + cloud).
- Directed a 60+ member cross-functional organization (software, hardware, QA/RA, clinical, PMO) to deliver high-quality product releases with >95% on-time delivery across a multi-year roadmap.
- Reduced product development cycle by 30% by implementing Agile transformation, improved backlog prioritization, and optimized portfolio governance.
- Solutions provided for long-term supplier and technology partnerships enabling IoT connectivity, SaMD capabilities, and field-update automation.
- Design, Prepare, Present and Monitor Dashboards for project Schedule & Cost variance and Risk management.
- Budget forecasting partnered with finance and procurement teams for vendor cost negotiation, SoW definition, CR Change order tracking, Capex/Opex approvals.
- Cybersecurity features for patient monitors to comply with security standards.(C, C++, VxWorks)
- Experience with cloud-hosted services and Azure DevOps.
- Integration to all major charting systems (EMR/EHR) via HL7, FHIR for interoperability (SaaS and SaMD).

COMPETENCIES

Strategic Leadership,
Team Development,
Communication,
Technical Excellence,
Process Optimisation,
Stakeholder Engagement,
Managing project Risks,
Innovation & Research,
Problem Solving,
Budget management,
Work for Team Success,
Conflict resolution

TECHNICAL PROFICIENCY

MS project, Office Timeline,
Agile Scrum, Waterfall,
MS Visio, MS 365, PowerBI.
Azure DevOps, CI/CD, DOORS,
IBM Synergy, PLM,
C, C++, C#, SQL, WCF, RESTful
API, Web service, .NET
Framework, Virtual machines,
Python,
VxWorks, Linux, Device
Drivers, RTOS. I2C, SPI, RS232,
IOT and Robotics, Manual &
Automation testing.
PowerPC, NXP, TI, Microchip,
ST, ESP32, PIC16/32, ARM.

EDUCATION/TRAININGS

Master of Computer
Applications-**MCA**.

Master of Business
Administration-**MBA (PM)**.

Trained PMP, AI and ML for
Healthcare, UX/UI Design.

COMPETENCIES

- Project execution for RoHS3 compliance, EU MDR compliance, Supply Chain challenges, and Obsolescence planning for Patient monitors.
- Close alignment with engineering, UX/UI, QA, marketing, and compliance teams.
- Adaptation to AI tools for smart project analytics, risk prediction, and resource optimization.
- Developed Capnography module (patent received), Multi GAS module, and OEM (ventilators and anaesthesia devices) interface modules, and developed Patient Monitor UI development in VxWorks.

Specialist Software Engineering, Rossell Techsys - 2015 to 2016

- As a Specialist Engineer, interacted with the Business development team and project management office and carried out requirements study, analysis, design and development of new products and sustaining products specific to Aerospace and Defence domain.
- Responding to technical proposals and involved in Air Borne Line replaceable unit (LRU) safety product clearance from CEMILAC, DGAQA agencies.

Principal Embedded Engineer, Spacelabs Healthcare - 2006 to 2014

- As a Principal engineer interacted with the marketing team and project management office and carried out requirements study, analysis, design and development of new products and sustaining products.
- As a Technical Lead responsibilities include Product Requirements, Software Development Planning, Software architecture design, proposals, Reviews and coordinating with cross functional teams for project planning and execution.
- Third party device interfaces, Capnography module (received Patent), Multi GAS module, OEM interface modules, Patient Monitor UI development.
- Patient simulator Design and development using PIC16x controller for vital sign parameters (ECG, IBP, SpO2, CO, Temperature and HR alarms).
- As an Individual contributor developed patient monitor User interface modules using Wind River Tilcon graphics suite for touch screen based patient monitors.
- Application software programming using VxWorks RTOS for interfacing between third party devices and Patient monitors on Power PC target platform.
- Maintenance projects: As a Lead involved in enhancing the product support team. Activities including Feature additions, Fix field issues and product enhancement and obsolescence porting

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Team Lead, Saranya Electronics Pvt.Ltd. 2004 to 2006

- Led team of engineers to build a custom Linux kernel for a single board computer.
- Acquisition modules firmware design and development (AT89c55, Keil51)
- Railway Platform coach position and Train status glance LED display system (AT89c55, Keil51)
- Firmware and PC based C# applications design and development for integration testing and production tools.

Project Engineer, HBL Nife Power Systems Ltd. 1999 to 2004

- Involved in getting product approval from Indian railways at Research Design and Standards Organization (RDSO)
- Implemented preventive maintenance algorithms for Railway signalling system as part of Data Logger Product development
- Involved and played key anchor roll in product demonstration and product installations
- Protocol design and development for network management using a MODBUS protocol. Interrupt driven routines using programmable interrupt controller 8259 and UARTs 8250.
- Menu driven user interface programming with 4 line LCD and key board
- Mentored team for developing Data logger GUI software using VB.NET and SQL Server
- Analog data acquisition & digital input data Acquisition logic
- Chronological storage of records using data structures
- Dot matrix printer interface and online print reports of digital, analog and alarm events
- Diagnostic routines for RAM, EPROM, Processor, Acquisition modules
- Serial communication in Visual Basic using MSComm control
- Live train movement and various signalling equipment in station yard monitoring .Net C#

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